

Daily Rainfall Data Analysis Report

Dataset Description

The dataset used in this analysis is the 'Daily Rainfall Data' file, which provides information about rainfall measurements across multiple locations. The CSV file contains records of daily rainfall amounts along with attributes such as Station Name, State, District, Date, and Rainfall (in mm). This dataset helps analyze rainfall distribution, seasonal patterns, and regional rainfall variations across different areas.

Key Columns include: Station Name, State, District, Date, and Rainfall (mm). The schema was inferred automatically using PySpark's DataFrame API.

Operations Performed

Data Loading & Schema Inspection

- Loaded the CSV file into a PySpark DataFrame using inferSchema.
- Displayed the first few rows and inspected column data types.
- Checked the total number of rows and columns present.

Exploratory Analysis

- Counted the total number of rainfall records.
- Verified unique stations, districts, and states present in the dataset.
- Displayed summary statistics such as minimum, maximum, mean, and standard deviation of rainfall values.

Aggregation and Grouping

- Computed total rainfall per state and district.
- Calculated average rainfall per station.
- Identified the top 10 rainiest districts and least rainy districts.

Data Quality and Null Value Checks

- Checked for null, zero, or missing rainfall entries.
- Verified any missing values in Station or Date columns.
- Filtered out invalid rainfall readings (negative or unrealistic values).

Correlation and Statistical Insights

- Computed correlations between rainfall and other numerical columns (if present).
- Derived summary statistics using mean, median, stddev, min, and max functions.
- Identified seasonal and monthly rainfall patterns using grouping and average aggregation.

Data Visualization

- Bar charts showing total rainfall per state and district.
- Line plots showing rainfall trends across time.
- Histogram illustrating the frequency distribution of rainfall values.
- Scatter plots showing relationship between rainfall and date/time for trend detection.

Key Insights

- The dataset covers multiple states and districts with daily rainfall records.
- Some regions experience significantly higher rainfall compared to others, indicating climatic diversity.
- The average rainfall varies seasonally, with peaks during monsoon months.
- A few stations have missing or zero rainfall entries which may indicate sensor or data recording issues.
- High rainfall districts include coastal and hill regions.
- Statistical summary shows variation with mean rainfall around moderate levels but large standard deviations in monsoon zones.

Recommendations

- Ensure consistent and complete data collection to handle missing rainfall values.
- Perform seasonal analysis (monthly/quarterly) to support agricultural and flood management planning.
- Integrate this dataset with temperature and humidity data for multi-factor climate analysis.
- Visualize rainfall data on maps for geospatial understanding of rainfall distribution.
- Investigate abnormal readings to improve data quality and reliability.

Example Snippet

Column Name	Description	Sample Value
Station_Name	Name of the rainfall measurement station	Pune Station
State	Name of the state	Maharashtra
District	Name of the district	Pune
Date	Observation date	2023-07-15
Rainfall	Recorded rainfall in millimeters	42.5